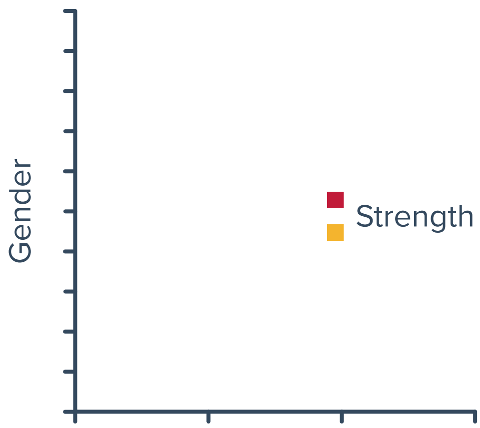
Independent and dependent variab

1 For each pairs of variables, state whether one variable has a direct effect on the other:

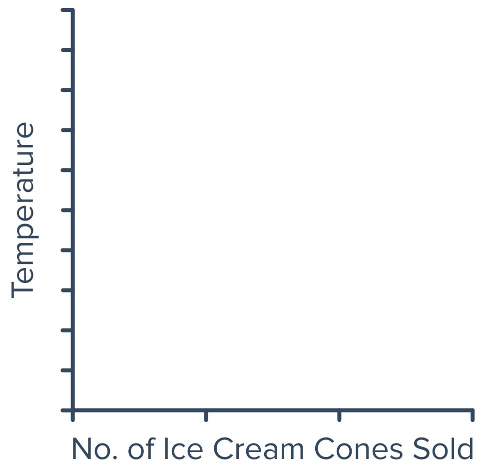
- **a** Arm length (cm), Weight (kg)
- **b** Temperature ($^{\circ}\mathrm{C}$), Distance from the equator (km)
- **c** Weekly income, Number of friends
- **d** Time of travel (minutes), Distance covered (m)
- **e** Age (years), Time spent watching television
- **f** Time spent working, Wages earned
- **g** Time spent to finish a novel, Number of pages
- **h** Marital status, Film preference
- **i** Height (cm), Number of children
- **j** Hair Length, Gender
- **k** Number of pets, Time spent caring for pets
- **l** Time spent training, Athletic performance

2 For the following sets of axes, state whether the variables have been placed in the correct position:

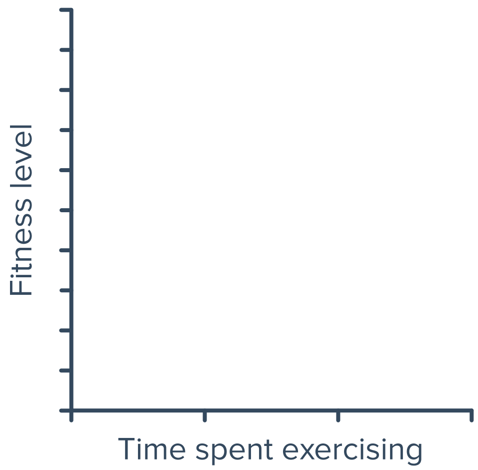
a



b



c



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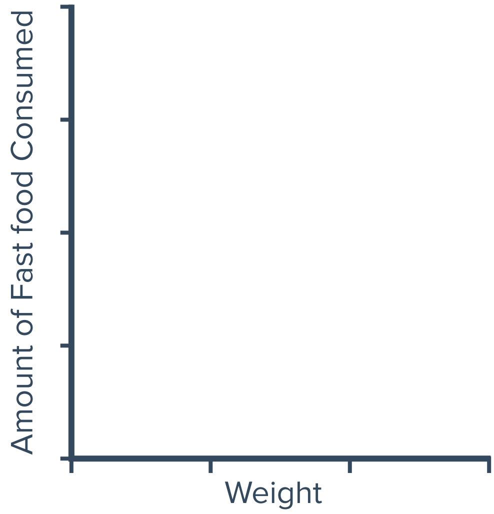


3 Consider the following variables:

- Ticket sales
- Revenue from a show

a State whether the following statements make sense:

i The revenue made from a show affects ticket sales.

ii Ticket sales affect the revenue made from a show.

b Which is the independent variable and which is the dependent variable?

4 For each of the following pairs of variables:

i State the independent variable. **ii** State the dependent variable.

a Number of ice cream cones sold and temperature outside.

b Size of a pizza and cost of a pizza.

c Time spent studying and exam mark.

5 A study was performed to find the relationship between the number of soft drinks consumed per week and the waist circumference of students.

a Which variable is the independent variable?

b Which variable is the dependent variable?

c Which variable should be put on the x-axis?

d Which variable should be put on the y-axis?

6 A kite can rise to a height of 161 ft when the wind is blowing at 6 miles per hour and to a height of 322 ft when the wind is blowing at 12 miles per hour.

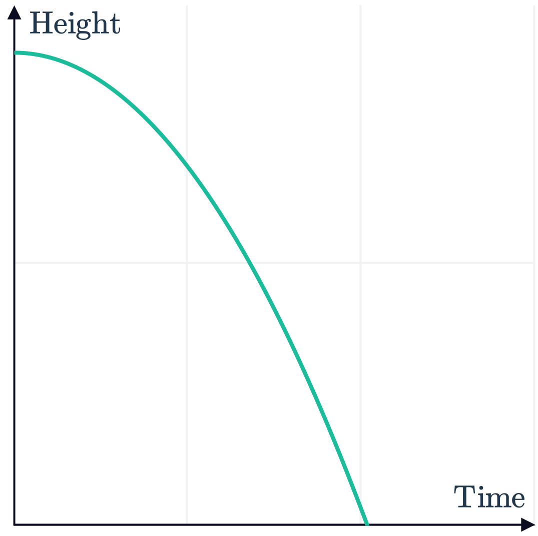
a What are the two variables in the above scenario?

b From your two variables, identify the dependent and independent variables.

7 The graph shows the height of a ball after it is dropped off the side of a building:

a Which variable is the dependent variable?

b Which variable is the independent variable?

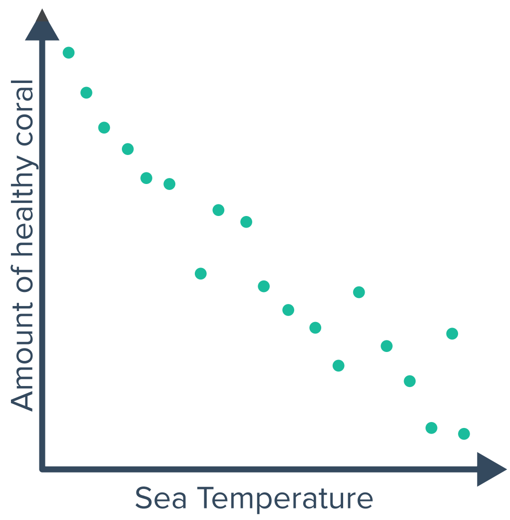


Scatter diagrams

8 The scatter diagram shows the relationship between sea temperature and the amount of healthy coral:

a Which variable is the dependent variable?

b Which variable is the independent variable?

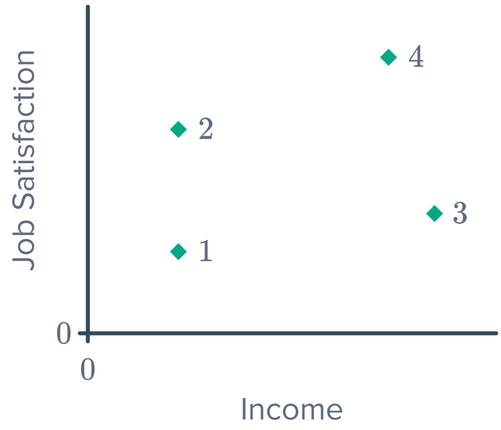


9 Construct a scatter diagram for the set of data in the following table:

x	1	3	5	7	9
y	3	7	11	15	19

10 Yvonne has a high income but does not lik her job.

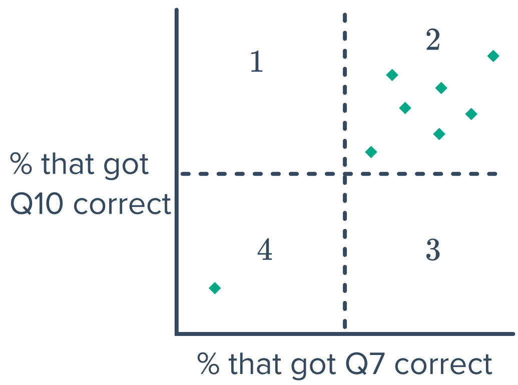
Which dot is most likely to represent Yvonne?



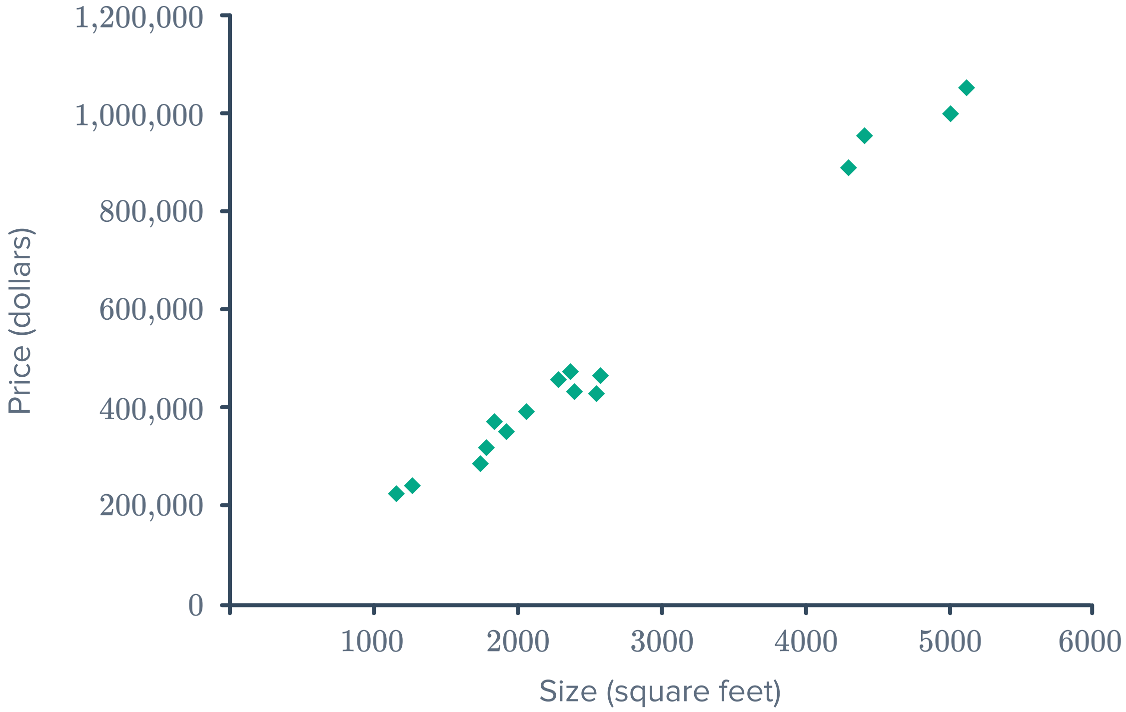
11 After a mathematics exam, Lachlan commented that he left out Question 10 because it was not relevant to the topics that were assessed, but that the other questions were fine. The teacher felt that Question 10 and Question 7 were assessing the same skills, so she decided to look at the results of students across all 8 classes. She plotted the percentage who had gotten Question 10 correct against the percentage who had gotten Question 7 correct as shown below:

a According to the graph, who is correct: Lachlan or the teacher?

b Let's assume that Lachlan performed exceptionally well in every other question. If a whole class of students had the same result as Lachlan, where could this class's results appear on the scatter diagram, region 1, 2, 3 or 4?



12 The land size of a sample of properties an[illegible]r selling price were graphed on the scatter diagram below:



a Which size range that contains the most properties?

A $4500 - 5500$ sq ft **B** $1500 - 2500$ sq ft **C** $2500 - 3500$ sq ft

b Another property that is not represented in the scatter diagram sold for $\$704\,000$ and measured 3600 sq ft in size. Is this consistent with the relationship shown in the graph?

13 Scientists were looking for a relationship b[redacted]en the number of hours of sleep we receive and the effect it has on our motor and process skills. Some subjects were asked to sleep for different amounts of time, and were all asked to undergo the same driving challenge in which their reaction time was measured. The data is recorded in the table below:

Amount of sleep (hours)	9	6	4	10	3	7	2	5
Reaction time (seconds)	3	3.3	3.5	3	3.7	3.2	3.85	3.55

a Construct a scatter diagram using the data from the table.

b Is amount of sleep and reaction time negatively correlated, postively correlated or not correlated?

c According to the results, does an increased amount of sleep generally lead to improved reaction times?

14 To determine whether the presence of sharks in a coastal region is influenced by cage diving, the number of nearby cage diving operations and the number of nearby shark sightings was recorded each month over several months. The results are shown in the table below:

Cage diving operations	10	2	4	7	8	5	1	9	6	3
Shark sightings	4	3	5	4	1	4	5	7	10	8

a Construct a scatter diagram using the data from the table.

b Describe the correlation between the number of shark sightings and the number of cage diving operations.

c According to the data, is it possible to determine whether cage diving operations encourage more sharks to come near the shoreline?

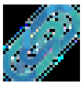
15 The market price of bananas varies throughout the year. Each month, a consumer group compared the average quantity of bananas supplied by each producer to the average market price (per unit).

Supply (kg)	Price (dollars)
550	15.25
600	14.75
650	14.75
700	14.75
750	14.25
800	14.00
850	13.75
900	13.25
950	13.50

a Construct a scatter diagram using the data from the table.

b Describe the relationship between the supply quantity and the market price of bananas.

c According to this data, when will a supplier of bananas receive a higher



Supply (kg)	Price (dollars)
1000	13.25